

Final Assignment: Completing an Entire Data Analytics Project

The main objective of this final assignment is to complete an entire data analytics project based on a given project **objective** and **scope**. In this assignment, you must work in groups to address specific **business questions** using a provided dataset and elements of the data analytics process.

Objective:

You are working in the Data Analytics department of Spotify and would like to understand more about the information available on song popularity, genre distribution, and audio characteristics. The objective of this project is to apply this knowledge to formulate customer-oriented strategies for both Spotify and its partner artists and production companies.

Scope:

You will choose one of the following three project tracks to address a corresponding set of business questions:

Track 1: Song Segmentation

- Can you identify distinct segments of songs based on their features?
- What are the key characteristics of each segment?
- How can marketing strategies be tailored to each segment to promote songs and increase listener engagement?

Track 2: Popularity Prediction

- Given a song's features, can you estimate its popularity score?
- What are the key features that influence a song's popularity on the platform?
- How can artists and producers use these insights to shape their musical decisions and increase the likelihood of success on the platform?

Track 3: Artist Recommendation

- Can you recommend artists whose musical style aligns with specific features found in a given song?
- What key characteristics imply an artist's overall musical style?

- How can these feature insights be used to identify and promote underrated artists with similar musical styles to more popular ones?

Data Overview:

The Spotify Tracks dataset contains approximately 125 music genres, with songs further categorized based on various audio features. It includes around 114,000 entries and 21 columns, capturing metadata and acoustic properties of each song. The dataset can be downloaded from the following link: [📁 Final Project Dataset](#) . It is sourced from Kaggle: [🎹 Spotify Tracks Dataset](#).

The features included in the dataset are described in detail below. You are not required to use all available attributes. Instead, you are encouraged to conduct feature selection and/or engineering to identify or create a subset of relevant features for the business problem you've chosen.

song_id: Unique identifier for the song in Spotify's database

artists: List of contributing artists' names, separated by semicolons if multiple

album_name: Title of the album containing the song

song_name: Title of the song

popularity: Numerical score ranging from 0 to 100, reflecting current listening metrics; derived algorithmically based predominantly on total play count and recency of plays. Songs with higher recent play counts have higher scores. Duplicate entries (e.g., same song from different sources) are rated independently. Artist and album popularity scores are mathematically derived from song popularity data.

duration_ms: Duration of the song expressed in milliseconds

explicit: Boolean indicator denoting presence of explicit content (true = explicit lyrics; false = non-explicit or unspecified)

danceability: Float value between 0.0 and 1.0 indicating suitability for dancing, based on musical attributes such as tempo, beat stability, rhythm regularity, and beat strength; where 0.0 is least danceable and 1.0 is most

energy: Float value from 0.0 to 1.0 representing perceived intensity and activity level; higher values denote fast, loud, and noisy songs (e.g., metal), while lower values signify calmer pieces (e.g., classical preludes)

key: Integer denoting the musical key, mapped to pitches via Pitch Class notation (0 = C, 1 = C $\sharp$ /D $\flat$, 2 = D, etc.); a value of -1 indicates no key detected

loudness: Numeric measurement of overall song loudness in decibels (dB)

mode: Categorical indicator of tonal modality; 1 = major mode, 0 = minor mode

speechiness: Float from 0.0 to 1.0 quantifying the presence of speech elements; values > 0.66 suggest predominantly speech-based content; between 0.33 and 0.66 suggest mixed music and speech; below 0.33 indicate predominantly musical content or non-speech audio

acousticness: Float from 0.0 to 1.0 estimating likelihood the song is acoustic; higher values reflect greater confidence in acoustic origin

instrumentalness: Float from 0.0 to 1.0 predicting the absence of vocal content; values approaching 1.0 imply the song is likely instrumental, with "ooh" and "aah" sounds treated as instrumental; songs with vocal content (e.g., rap, spoken word) tend to have lower scores

liveness: Float from 0.0 to 1.0 indicating the probability that the recording was performed live; values > 0.8 strongly suggest a live performance

valence: Float from 0.0 to 1.0 measuring musical positiveness; higher scores denote happier, more cheerful songs, while lower scores indicate sadness, depression, or anger

tempo: Estimated beats per minute (BPM) reflecting the song's overall pace; derived from average beat duration

time_signature: Integer estimate of the musical time signature, ranging from 3 to 7, representing time signatures from 3/4 to 7/4

song_genre: Assigned genre category of the song

Instructions:

1. Discuss your ideas with your group members.
2. Select one of the provided three project tracks to center your project around
3. Download the dataset: [Final Project Dataset](#)
4. Prepare the dataset
5. Conduct basic data analysis to generate descriptive statistics and identify any patterns or trends in the data
6. Apply the most appropriate statistical modeling to make predictions and address the project's objective
 - Consider using the K-means clustering algorithm to segment songs based on their features
 - Consider using supervised learning algorithms such as Regressions, Decision Trees, and Random Forests for predictive tasks
7. Interpret the findings to generate insights and recommendations that answer the business question(s)
 - Consider interpreting the characteristics of each song segment (e.g., high-energy dance songs, acoustic songs with low tempo, etc.) and suggest tailored engagement strategies for each song segment. For instance, you may propose curated playlists or targeted campaigns to promote underrepresented song types.
 - Consider the features that you've chosen or engineered are most important for predicting a song's popularity, and formulate artist- or platform-level strategies around those insights. For example, if songs with higher danceability and tempo

are strongly associated with higher popularity scores, you may suggest that artists incorporate these traits to improve reach, or that Spotify prioritize promoting such songs to boost engagement.

8. Communicate the results in a 5-min recorded presentation, including the following elements:
 - Contributors
 - List the members of your team. Include names, school, degree of study, and current job (if applicable)
 - Defining the problem
 - Explain the objective of the data analytics project, and (re)state the problem(s) and/or the business question(s) to be addressed
 - Data collection and preparation
 - Briefly describe the data used for the analysis and outline the data preparation process
 - Data exploration
 - Briefly discuss how initial data exploration was conducted
 - Did you identify any patterns or trends in the data? Share descriptive analytics and any initial insights.
 - Application of statistical and machine learning techniques
 - Talk about the modeling approach(es) and techniques used for the analysis
 - Which model(s) worked best for the project's objectives and why?
 - Interpretation of results
 - Share key insights from the analysis and discuss how the findings can help solve the problem(s) and/or answer the business question(s)
 - What is the proposed solution(s) and/or recommendation(s)?
 - Communicating the results
 - When presenting the data analytics project, communicate the insights and findings (to stakeholders) in a way that is easily understood and actionable
 - Be sure to use data visualizations to help with that communication!
8. Share the link to your presentation in the [Final Project Peer Review](#) discussion board. Watch and comment on your peers' presentations in order to receive **extra credit**.

How to record the video:

- If you have worked with video-recording software in the past, please feel free to use any tool you are familiar with
- Please see the [instructions for recording screen on an Apple device](#)
- If you have a PC, you can use [this free Chrome extension](#)
- You may also use Zoom, and download the recording afterwards
- As long as you have clearly documented all of the aspects of your project listed in the rubric, you will not be graded on the production quality of your video.
- Make sure to demonstrate each component of the rubric to receive full credit.
- Please upload the document and video as separate files. Multiple file uploads are enabled for this assignment
- Please watch your video before you submit it to ensure it functions properly and fully demonstrates your project. Incomplete videos may be subject to lost rubric points.

Deliverables:

Submit all code used for data cleaning, exploration, and modeling, along with any Tableau project files used for visualization; the slide deck used for the presentation; the 5-min video recording; and post the recording link in the Final Project Peer Review discussion board.

If you'd like to share your thoughts on working with your teammates, please use the [Project Contribution form](#). The form is completely optional, and will be taken into consideration when grading the Final Project.

Grading Criteria:Project Completeness (20/100):

The project addresses all business questions of the corresponding project track on the basis of completion.

- Track 1:
 - Implements a model that identifies distinct song segments based on their features (10pts)
 - Analyzes key characteristics of each song segment (5pts)
 - Provides at least one marketing strategy that applies characteristics of song segments (5pts)
- Track 2:

- Implements a model for estimating popularity scores based on its features (10pts)
- Analyzes the model to determine which features have the greatest influence on a song's popularity (5pts)
- Provides at least one strategy that applies insights from the model to guide music production decisions based on song-level popularity trends (5pts)
- Track 3:
 - Implements a model that recommends artists based on the features of a given song (10pts)
 - Analyzes the model to determine which features most influence artist similarity and musical style (5pts)
 - Provides at least one strategy that applies insights from the model to identify and promote underrated artists with similar musical styles to more popular ones (5pts)

Code Quality (10/100):

Project codebase should be clear and well-organized with concise comments, well-named variables, and reusable components.

- Readability and commenting: 5pts
- Modularity and organization: 5pts

Functionality and Performance (30/100):

Project methodology should demonstrate well-motivated data exploration and effectiveness of the implemented models.

- Correctness of data exploration: 10pts
 - Necessary data preprocessing is properly conducted (5pts)
 - Features chosen for model implementation are well-motivated (5pts)
- Correctness of models/logic: 20pts
 - Chosen model family is suitable for addressing problem statement (5pts)
 - Implementation of model training and inference functions is correct (5pts)
 - Metrics for evaluation of model performance are suitably chosen and correctly implemented (5pts)
 - Model performance is reasonable and reproducible (5pts)

Presentation (40/100):

The presentation should effectively communicate the project's objectives, methodologies, and findings.

- Clearly lists contributors: 5pts
- Clear definition of objective: 5pts
- Clear description of data collection and preparation process: 5pts
- Comprehensive data exploration: 10pts
- Insightful interpretation of results and presentation of business strategies: 10pts
- Overall clarity of presentation and visualizations: 5pts